

How the *spike* number is made

One belief per cell, where a cell is one subject crossed with one way of working. Every constant read from the running engine; every figure computed by it.

WHAT MOVES IT	WEIGHT, BEFORE DECAY
cross_day_return	α +1.0
The latest engagement of this cell before this one falls on an earlier UTC day, and we did not nudge them here. Only the most recent predecessor counts.	
unrequired_revision · chosen_challenge failure_recovery · self_authored_scope	α +0.5
Did something harder than they had to. Presence counts; strength is never read.	
external_report · broad	α +0.25
A named adult saw the interest reach the topic itself, not the materials. Never marks a day.	
skip	β +0.5 ÷ choice set
On offer, not taken, and they have engaged this cell before.	
decline	β +0.15 ÷ choice set
On offer, not taken, never engaged. You cannot tire of what you never tried.	

WHAT SCORES NOTHING	
same_day_engagement	0
A first-ever touch, a reopen in one session, or a re-entry the same day. Has not survived a night away.	
prompted_return	0
We surfaced or recommended them into it. Proves we recommend well.	
novelty (a flag on any event)	0
Within 3 days of first exposure to the cell. Triggered situational interest, excluded both ways.	
artifact_competence · report focused	0
How good the thing is, and interest that stayed with the materials. Different questions.	
dwelt · attempts · work quality	0
Recorded on the raw interaction and structurally absent from the type the belief math reads, so they cannot be multiplied in.	
candidate → consecutive returns 14 days apart · 2 cycles over 56 days · made something that survived a setback enough. Observed high as clicks happen.	

THE ARITHMETIC
$\alpha_0 = 1 + .5 \cdot env + .5 \cdot mastery + .5 \cdot disc$
$\beta_0 = 1 \quad // \text{ nothing tilts against}$
$r = 0.5 ^ { \left(\text{days} \div 14 \right) }$
$\alpha = \alpha_0 + \sum \left(w \cdot role \cdot r \right)$
$\beta = \beta_0 + \sum \left(w \cdot r \div \max(1, \text{set}) \right)$
$n = \alpha + \beta$
$mean = \alpha \div n$
$var = \alpha \beta \div \left(n \cdot n \cdot (n+1) \right)$
LB = mean - $\sqrt{var}$ // the output

The lower bound, not the mean. Two events and twenty events can share a mean of 0.85. Subtracting a standard deviation separates them, so thin evidence is penalised automatically.

RECENCY	HALF-LIFE 14 DAYS					
	0d	7d	14d	28d	42d	56d
	1.00	0.71	0.50	0.25	0.13	0.06
Two masses. α and β decay and answer <i>what do we believe now</i> . observedMass sums the same weights undecayed and answers <i>have we seen enough</i> . Observed high as clicks happen.						

ALL FOUR MUST HOLD	
observedMass ≥ 6 Undecayed. One return banks per day, so about a week.	distinctDays ≥ 2 A count runs up in one sitting. A calendar cannot.
2·sd ≤ 0.35 The band has to be narrow.	LB ≥ 0.6 No evidence sits at 0.5.

WHY FOUR AND NOT ONE	ENGINE-COMPUTED		
SAME CHILD, DIFFERENT WEEKS	LB	MASS	DAYS
Six returns across 15 days	0.70	6.0	6 ✓
... plus ten skips, 7-way choice	0.62	6.7	13 ✓
Six returns, one single day	0.76	6.0	1 ✗
Three returns, three days	0.62	3.0	3 ✗

Rows three and four both beat the 0.6 threshold and are still refused. A high number alone is never enough.

One choice moment costs one skip. The divisor is the count of cells offered, not taken and past novelty, so the six things passed over to pick a seventh share a single skip's worth between them. Without it, breadth of discovery would bury the spike: those same ten skips would drive the second row from 0.62 down to **0.38**.
THE ENGINE NEVER PROMOTES
SYNTHETIC DATA · NO REAL CHILD